Adding Numbers in Scientific Notation

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Scientific Notation

Scientific notation expresses a number in the form

$$a \times 10^n$$

where $1 \leq |a| < 10$ and n is an integer.

Adding Numbers

To **add numbers in scientific notation**, their powers of 10 should have the **same exponent** before adding their coefficients.

$$2 \times 10^5 + 3 \times 10^5$$

Same exponent: 10^5

Different Exponents?

If the exponents are different, **rewrite** one or both numbers so that they have the same power of 10.

$$3.5 \times 10^5 + 2.8 \times 10^4$$

The exponents are 5 and 4.

Rewriting a Number

To raise the exponent by 1, move the decimal point one place to the left.

$$2.8 \times 10^4 = 0.28 \times 10^5$$

Now both numbers have 10^5.

Add the Coefficients

Once the exponents are equal, **add the coefficients** and keep the common power of 10.

$$(a + b) \times 10^n$$

Normalize the Answer

If the resulting coefficient is

greater than or equal to 10, or less than 1,

normalize the answer by adjusting the coefficient and exponent.

Normalizing: Example

$$13.7 \times 10^4 = 1.37 \times 10^5$$

The decimal point moved 1 place to the left, so the exponent increased by 1.

Final Answer

Always express the final answer in proper **scientific notation**.

$$1 \leq |a| < 10$$

Steps to Remember:

SAME

S: Standardize

S – Standardize the exponents by rewriting the numbers.

A: Add

A – **Add** the coefficients.

M: Maintain

M – Maintain the common power of 10.

E: Ensure

E – Ensure that the answer is in proper scientific notation.

SAME Summary

S **S**tandardize the exponents
A **A**dd the coefficients
M **M**aintain the power of 10
E **E**nsure proper scientific notation

Examples

Example 1

Add.

$$2.4 \times 10^5 + 3.1 \times 10^5$$

The exponents are the same, so add the coefficients.

$$(2.4 + 3.1) \times 10^5 = 5.5 \times 10^5$$

Example 2: Standardize

Add.

$$3.5 \times 10^5 + 2.8 \times 10^4$$

S: Rewrite 2.8×10^4.

$$2.8 \times 10^4 = 0.28 \times 10^5$$

Example 2: Add and Maintain

$$3.5 \times 10^5 + 0.28 \times 10^5$$

A: Add the coefficients.
M: Keep 10^5.

$$(3.5 + 0.28) \times 10^5 = 3.78 \times 10^5$$

Example 2: Ensure

E: Is 3.78×10^5 in proper scientific notation?

$$1 \leq 3.78 < 10$$

Yes!

Example 3

Add.

$$7.5 \times 10^4 + 6.2 \times 10^4$$

$$(7.5 + 6.2) \times 10^4 = 13.7 \times 10^4$$

13.7 is not less than 10, so normalize.

$$13.7 \times 10^4 = 1.37 \times 10^5$$

Example 4

Add.

$$2.3 \times 10^{-4} + 4.6 \times 10^{-5}$$

S: Rewrite 4.6×10^{-5}.

$$4.6 \times 10^{-5} = 0.46 \times 10^{-4}$$

Example 4: Finish

$$2.3 \times 10^{-4} + 0.46 \times 10^{-4}$$

$$(2.3 + 0.46) \times 10^{-4} = 2.76 \times 10^{-4}$$

Proper scientific notation: $1 \leq 2.76 < 10$

Practice Exercises

Practice 1

Add the following numbers expressed in scientific notation.

$$2.4 \times 10^6 + 3.1 \times 10^6$$

Practice 2

Add the following numbers expressed in scientific notation.

$$5.2 \times 10^4 + 2.6 \times 10^4$$

Practice 3

Add the following numbers expressed in scientific notation.

$$1.7 \times 10^8 + 4.2 \times 10^8$$

Practice 4

Add the following numbers expressed in scientific notation.

$$3.5 \times 10^5 + 2.8 \times 10^4$$

Practice 5

Add the following numbers expressed in scientific notation.

$$6.4 \times 10^7 + 1.3 \times 10^6$$

Practice 6

Add the following numbers expressed in scientific notation.

$$4.6 \times 10^3 + 5.7 \times 10^2$$

Practice 7

Add the following numbers expressed in scientific notation.

$$7.8 \times 10^{-3} + 1.5 \times 10^{-3}$$

Practice 8

Add the following numbers expressed in scientific notation.

$$2.3 \times 10^{-4} + 4.6 \times 10^{-5}$$

Practice 9

Add the following numbers expressed in scientific notation.

$$8.2 \times 10^9 + 3.5 \times 10^8$$

Practice 10

Add the following numbers expressed in scientific notation.

$$6.7 \times 10^5 + 4.1 \times 10^5$$

Activities

Activity 1

Add the following numbers expressed in scientific notation.

$$3.2 \times 10^6 + 4.5 \times 10^6$$

Activity 2

Add the following numbers expressed in scientific notation.

$$4.3 \times 10^4 + 3.5 \times 10^4$$

Activity 3

Add the following numbers expressed in scientific notation.

$$2.6 \times 10^8 + 5.1 \times 10^8$$

Activity 4

Add the following numbers expressed in scientific notation.

$$4.2 \times 10^5 + 3.6 \times 10^4$$

Activity 5

Add the following numbers expressed in scientific notation.

$$5.3 \times 10^7 + 2.4 \times 10^6$$

Activity 6

Add the following numbers expressed in scientific notation.

$$3.8 \times 10^3 + 4.9 \times 10^2$$

Activity 7

Add the following numbers expressed in scientific notation.

$$6.4 \times 10^{-3} + 2.7 \times 10^{-3}$$

Activity 8

Add the following numbers expressed in scientific notation.

$$3.5 \times 10^{-4} + 2.8 \times 10^{-5}$$

Activity 9

Add the following numbers expressed in scientific notation.

$$7.4 \times 10^9 + 4.2 \times 10^8$$

Activity 10

Add the following numbers expressed in scientific notation.

$$5.8 \times 10^5 + 3.6 \times 10^5$$

Thank you!